

What Carries Deep-Noise Robustness in a Mamba-3 Block? Input-Independent Scans and Noise-Dependent Gating under Cross-Condition Shift

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Abstract

The robustness of Mamba-style selective state-space models is often attributed to input-dependent selectivity. We test this attribution with pre-specified ablations of one Mamba-3 block on bearing vibration data under cross-condition shift (XJTU-SY, one speed-and-load pair, training and evaluation bearings disjoint) with matched-noise training, reporting macro-F1 at the final epoch of a fixed budget. The central contrasts were re-run with training and evaluation noise drawn independently. Making the scan’s dynamics and read/write coefficients input-independent (Δ , A , B , C , trapezoidal weight and rotation angles) does not remove the block’s advantage over a time-invariant S4D: on the original transition the fully frozen block leads by +21.3, +23.3, +20.3 pp at 0, -2 and -6 dB (all intervals excluding zero). Removing the multiplicative output gate costs +9.4, +15.7, +27.0 pp in the partially frozen block and +5.4, +9.3, +32.6 pp in the fully frozen block, resolved only at -6 dB (on the reverse transition, at every level); without its gate the fully frozen block still exceeds S4D at 0 and -2 dB but falls below it at -6 dB. Replacing the gate by an additive branch of identical size and initial weights leaves the result indistinguishable from removing it. Adding a gate to S4D recovers part of the gap, no more than an additive branch of the same size on one transition direction and slightly more on the other. On Paderborn data the gate-removal cost replicates when training and evaluation share bearings, the freeze advantage only at -6 dB, and nothing replicates on disjoint bearings, where every architecture transfers poorly.

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The evidence supports input-independent scan coefficients sufficing in this block and a gate whose necessity grows with noise depth; it is limited to one block, one shift pair per dataset and five seeds.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells of the original grids (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it. The chain has three steps, each with adjudication thresholds fixed before the corresponding grid was run:

1. **Test the classical selectivity terms (necessity test).** We construct a *frozen* arm: the input-dependent parts of Δ , A , B and C are replaced by learned constants. The Mamba-3 scan also takes an input-dependent trapezoidal weight and input-dependent rotation an-

gles; these, the convolutional stem, the multiplicative SiLU gate, the state parameterisation, normalisation and projections are retained (Section 3.2). If these four terms drive robustness, the frozen arm should collapse toward the time-invariant baseline.

2. **Localise the necessary component.** On the frozen base, we remove one component at a time (convolutional stem; SiLU gate; heavy-tailed state parameterisation) under a single-variable discipline checked key-by-key on the state dictionary (Section 3.2), and adjudicate each arm against pre-specified Δ -thresholds.

3. **Test sufficiency by constructive grafting.** The component identified as necessary is added to plain S4D as a gate branch, and a pre-specified recovery ratio decides whether “removal collapses” is matched by “addition restores”.

The outcome is a conditional pattern. Under independent training and evaluation noise (Section 4.1), making the scan’s dynamics and read/write coefficients input-independent does not remove the block’s advantage over S4D (+21.3, +23.3, +20.3 pp for the fully frozen block). Removing the gate costs +9.4, +15.7, +27.0 pp in the partially frozen block, resolved only at −6 dB, and without its gate the fully frozen block still exceeds S4D at 0 and −2 dB but not at −6 dB: the gate’s contribution grows with noise depth. Replacing the gate by an additive branch of identical size cannot be distinguished from removing it, in either host. Adding a gate to S4D, rebuilt in the donor’s width, recovers $R = 0.30$ of the gap (paired-seed interval 0.14–0.72), not more than an additive branch with identical parameters (0.41) on the original transition, and somewhat more on the reverse one (Sections 4.6 and 4.8). On Paderborn the gate-removal cost replicates when training and evaluation share bearings and nothing replicates on disjoint bearings, where no architecture transfers well (Section 4.9). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under operating-condition shift and matched noise, from the classical selectivity terms (Δ , A , B , C) to a scan whose dynamics and read/write coefficients are all input-independent, in which the block’s deep-noise advantage over S4D survives every step, including under independent training and evaluation noise; each arm’s frozen coefficients are verified input-independent at the kernel interface. (ii) Same-host and cross-host tests of the output

67 gate: removal, in-place replacement by an additive branch of identical size,
68 and grafting onto S4D with an additive control, showing a gate contribution
69 that is resolved at the deepest noise level and an additive branch that does
70 not substitute for it, while a time-invariant host gains little that is specific
71 to gating. (iii) A correction of the noise protocol (training and evaluation
72 noise originally shared by index) with a pre-specified re-run, and final-epoch
73 reporting under a fixed budget with best-epoch selection as a sensitivity
74 analysis. (iv) A documented pre-specification and review trail, including the
75 incidents in which automated adjudication departed from the pre-specified
76 criteria; we release the trail.

77 2. Background and Related Work

78 2.1. Time-invariant and selective SSMs

79 Diagonal structured SSMs (S4D) [6] evolve a linear state with input-
80 independent parameters (A , B , C , Δ), acting as a bank of learned linear
81 filters. Mamba-style blocks [1] make Δ , B , C functions of the input token
82 (selectivity), wrap the scan in a block containing a short causal convolu-
83 tional stem, a multiplicative SiLU gate on a parallel branch, and residu-
84 al/normalisation structure; the concrete block used throughout this paper
85 follows the Mamba-3 formulation [7]. The narrative crediting selectivity for
86 robustness draws on theory showing input-dependent dynamics can represent
87 discontinuity-sensitive projections and counteract memory decay [8], and on
88 [9]. We treat these as *hypotheses to be tested*, not established attribution-
89 s: the theoretical results concern representability, which does not by itself
90 determine which component earns empirical robustness in a trained block.

91 2.2. The unresolved comparison

92 Empirically the picture is contested. A systematic comparison on 59 clean
93 multivariate time-series classification benchmarks (MONSTER and UEA)
94 finds S4D-based models superior to Mamba variants [4]; that study does not
95 include additive noise or a shift of operating condition between training and
96 evaluation, which is the regime examined here; theoretical work identifies
97 long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, appli-
98 cation literatures—including machine-condition monitoring—contain many
99 Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but
100 none, to our knowledge, isolates *which component* of the block is responsible

under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.11). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.11 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every cell on the original transition carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication; the reverse transition and the two Paderborn settings carry their own label hashes (93424834..., 53d565d9... disjoint, addc1938... shared). The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475;

136 frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and
 137 follow-up grids add 645 cells (Sections 4.1, 4.3, 4.10, 4.6, 4.7, 4.8 and 4.9).
 138 Training and evaluation bearings are disjoint in each direction: Bearing2_1,
 139 2_2, 2_4, 2_5 at 37.5 Hz/11 kN and Bearing3_1, 3_3, 3_4, 3_5 at 40 Hz/10 kN;
 140 the reverse transition swaps their roles.

141 **Composition of the data.** Training uses the bearings of the 37.5 Hz/11 kN
 142 condition whose failure mode is inner race or outer race (Bearing2_1 is la-
 143 belled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of
 144 the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5
 145 are OR). Bearings with cage or mixed failure modes are excluded. Windows
 146 are non-overlapping segments of 2048 samples of the horizontal channel taken
 147 from the fault phase, which is delimited by a kurtosis/RMS onset criterion
 148 with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR
 149 windows, the training set 4,624 OR and 624 IR windows. Two consequences
 150 matter for interpretation. The IR class is learned from a single training bear-
 151 ing, so class-level results may reflect bearing-specific generalisation. And the
 152 five seeds resample initialisation, batch order and noise but not bearings:
 153 there are four independent bearings on each side, and none of the intervals
 154 reported here captures bearing-level uncertainty. One follow-up grid (Sec-
 155 tion 4.8) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the
 156 same eight bearings with their roles swapped, so it tests the other direction
 157 rather than independent bearings; its evaluation set holds 4,624 OR and
 158 624 IR windows, all IR windows coming from Bearing2_1, and its label hash
 159 is 93424834 The 35 Hz/12 kN condition has no inner-race bearing and
 160 cannot form a binary pair.

161 The principal arms are summarised in Table 1.

162 3.2. *What the frozen arm freezes, and what the checks verify*

163 In the Mamba-3 layer used here, one input projection emits the gate and
 164 value streams, the scan inputs B and C , the step-size and decay terms (Δ
 165 and A , through input-dependent additive terms), a trapezoidal weight and
 166 rotation angles. The frozen arm removes the input-dependent parts of Δ ,
 167 A , B and C and replaces them with learned constants broadcast over time
 168 (for Δ the static bias of the original layer is kept; for A a new constant pa-
 169 rameter plays the role of a static log-decay, passed through the layer’s own
 170 heavy-tailed clamp). The removed projections are deleted, so no parameter
 171 or gradient can leak. **The trapezoidal weight and the rotation angles**
 172 **remain functions of the current input**, as do the stem and the output

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.4)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.4)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.4)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: scan dynamics and read/write coefficients input-independent (value stream and gate still input-dependent)	103,842
frozen-all-gate	frozen-all with the gate rows removed from the input projection (Section 4.1)	71,074
frozen-all+add	frozen-all with $y \odot \text{SiLU}(z)$ replaced by $y + \text{SiLU}(z)$; same parameters and initial weights	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.5)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.6)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules, parameters and initial weights	112,410

gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3 formulation [7] the input-dependent rotation and trapezoidal weighting also enter the state update. Section 4.2 is accordingly a statement about the classical selectivity terms Δ, A, B, C only. Three further arms close that gap by the same device: *frozen-trap* replaces the trapezoidal weight by a learned per-head constant, *frozen-rot* replaces the rotation angles by learned constants (so the rotation becomes a learned position-only phase rather than a function of the current input), and *frozen-all* does both, so that the scan’s dynamics and read/write coefficients (Δ, A, B, C , trapezoidal weight, rotation angles) no longer depend on the input. The value stream x and the gate input z are still computed from the input, as they must be for the layer to carry any signal; the rotation becomes a learned position-dependent phase. In each case the corresponding `in_proj` output slice is deleted. Section 4.3 reports them. For these arms the construction check adds the decisive probe: the tensors handed to the scan are captured for two different inputs through the same module, and a tensor the arm claims to have frozen is required to be bitwise identical across them while a tensor left selective is required to differ (all three arms pass). No kernel modification is needed, because the scan interface accepts these tensors regardless of their origin.

Single-variable discipline was checked by key-level state-dictionary diffs between two models built under the same seed. For the frozen arm the diff verifies that every non-target tensor has the same name and shape as in BM3; for the component-ablation and graft arms it additionally compares the values of non-target tensors at initialisation. Every differing key is mapped to an entry in an architecture map. We do not claim byte-level identity of non-target tensors in the frozen arm, and trained weights naturally differ between arms; the checks concern how the arms were constructed.

3.3. Reporting rule

Every cell is trained for a fixed budget of 50 epochs with a cosine learning-rate schedule that anneals to zero, and the macro-F1 reported in the main text is that of the *final epoch*, a rule that uses no evaluation data to choose a model. The original grids recorded instead the maximum over epochs of macro-F1 on the evaluation split (the harness defines no separate validation split), and the thresholds pre-specified for those grids were written against that rule. Every arm of the chain was therefore re-run on the adjudication band (0, −2, −6 dB) with the original training code, the same evaluation

209 fingerprint and per-epoch evaluation (Section 4.10). Adjudications are re-
 210 ported under the rule they were specified for and under the final-epoch rule,
 211 and best-epoch values are given as a sensitivity analysis. Levels outside the
 212 band (clean, +10, +6, −10 dB) and the per-class grid were not re-run; values
 213 there are best-epoch and labelled auxiliary. Because the final epoch of the
 214 same evaluation condition has been inspected repeatedly during this study,
 215 final-epoch values are not a fresh confirmatory test either.

216 3.4. Confound elimination

217 Before component attribution, three candidate confounds were examined
 218 on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D
 219 (S4D-wide − S4D: −0.4, −1.0, +0.4 pp at 0, −2, −6 dB under final-epoch
 220 reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide
 221 by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so
 222 this particular increase in S4D’s parameter count does not explain the gap;
 223 the control does not address the change in channel mixing that gate removal
 224 also entails. **Normalisation:** both SSM arms use LayerNorm and contain
 225 no BatchNorm; the arm that degrades under noise is the one *without* any
 226 batch statistics, which argues against BatchNorm statistics mismatch as the
 227 explanation. Other normalisation inside the Mamba-3 layer was not varied.
 228 **Noise streams:** in the original harness the noise added to window i was
 229 drawn from a generator keyed on (seed, i), and training and evaluation used
 230 the same seed, so window i of the training set and window i of the evaluation
 231 set carried the same base noise vector. Per-index label agreement between
 232 the two sets is 0.549 on XJTU-SY, below the 0.628 expected by chance, so a
 233 model that memorised training noise would be pushed *away* from the correct
 234 test label; on Paderborn it is 0.999 (disjoint) and 1.000 (shared). All central
 235 contrasts were therefore re-run with the stream keyed on (seed, split, i), which
 236 makes training and evaluation noise independent while all arms with a given
 237 seed share one evaluation noise bank (Section 4.1); the reverse transition was
 238 re-run in the same way (Section 4.8). The component ablations of the stem
 239 and the state parameterisation, the grafts and the per-class grid were not
 240 re-run and carry the earlier protocol in their captions. For these, the shared
 241 noise is unlikely to favour the gated arms: on XJTU-SY the per-index label
 242 alignment between training and evaluation runs against the correct label,
 243 and in every arm that was re-run the means moved by at most 4 pp with
 244 no contrast changing direction. **Auxiliary physics loss:** an exploratory
 245 kinematics-alignment regulariser carried by the original model contributes

246 -0.1 pp pointwise (curves coincide); all attribution arms are therefore built
247 on the pure cross-entropy variant.

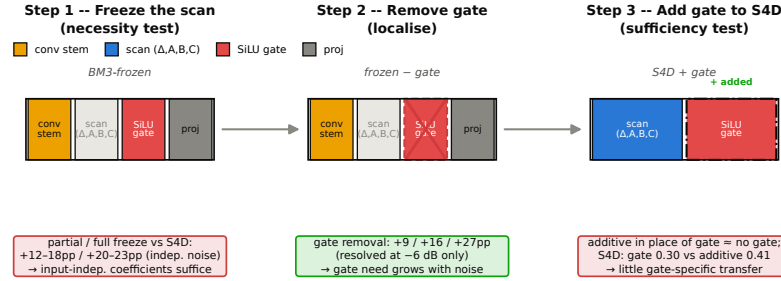
248 3.5. *Pre-specification and adjudication*

249 Every adjudicated question in this study had numeric decision thresholds
250 written to a task card and to disk before the corresponding grid was launched.
251 This is internal pre-specification, not external registration: the supporting
252 evidence is local file modification times and run-directory names (released,
253 [Appendix A](#)), which are not tamper-evident and are not preserved by a
254 repository checkout, so the trail lets readers inspect the order of events but
255 cannot prove it. The exception is the last three follow-up grids (Sections 4.6–
256 4.10): their thresholds, and for the last two grids the title decision that
257 depends on their outcome, were committed to the public repository before
258 the first cell was run, which gives those grids an external time-stamp. Later
259 questions (component ablation, graft) were specified after earlier grids had
260 been seen, as is natural in a staged design. Adjudication was mechanical:
261 an analysis pass applies the thresholds and emits a decision artifact; an
262 independent reviewer pass re-computes aggregates, verifies fingerprints row-
263 by-row, and must cite file paths and modification times for every check. The
264 pre-specified thresholds were binding in practice: two attempted post-hoc
265 criterion substitutions and one stale-artifact false verification were caught
266 and re-adjudicated under the original rules — this is the one place where pre-
267 specification actually bit. [Appendix C](#) details the incidents, the automated
268 executor/reviewer implementation, and the human adjudication gates; all
269 scientific decisions — thresholds, rulings, and interpretations — were made
270 by the authors and are timestamped. Statistical tests (paired, per-condition)
271 are reported throughout but were never adjudication criteria. A single repeat
272 of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-
273 F1 (non-deterministic kernel reductions; [Appendix B](#)); this is an observed
274 repeat difference, not an estimated variability bound, and arm-level contrasts
275 are judged against seed-level variability (2–9 pp).

276 4. Results

277 Throughout, “deep noise” denotes the pre-specified adjudication band
278 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
279 invariant baseline is near chance there: a constant majority-class predictor
280 scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this

281 evaluation set, against 52.3% for S4D at the best epoch). Section 4.1 reports
 282 the central contrasts under independent training and evaluation noise and is
 283 primary for the arms it contains; the later subsections report the full chain
 284 under the earlier noise protocol (training and evaluation noise shared by
 285 index, Section 3.4), labelled as such. All values are macro-F1 (%), mean
 286 over 5 seeds, at the final epoch of the 50-epoch budget (Section 3.3); best-
 287 epoch values are given alongside or in Section 4.10, and values outside the
 288 adjudication band are best-epoch and auxiliary. Figure 1 summarises the
 289 three-step ablation chain that structures this section.



Conditional on this block, testbed and budget: input-independent scan coefficients suffice; the gate matters most in the deepest noise.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.2), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.4), and graft that ingredient onto S4D to test sufficiency (Section 4.5).

290 4.1. Central contrasts under independent training and evaluation noise

291 A pre-specified grid of 150 cells (thresholds, labels and writing conse-
 292 quences committed to the public repository before the first cell) re-ran the
 293 central arms with independent training and evaluation noise (Section 3.4);
 294 everything else is unchanged. Each contrast is labelled *holds* if at least two
 295 of the three levels have a positive mean with an interval excluding zero and
 296 none is reversed, *weakened* if exactly one, *does not hold* if none. Across the
 297 eight re-run arms, the arm means moved by -3.4 to $+3.9$ pp relative to the
 298 earlier protocol (Table 2); no contrast changed direction.

299 *Freezing.* Both freeze contrasts hold. The fully frozen block exceeds S4D
 300 by $+21.3$, $+23.3$, $+20.3$ pp, in every seed at every level, with the smallest

Table 2: Independent training and evaluation noise, original XJTU-SY transition. Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds. Lower block: paired differences with 95% t -intervals and the number of seeds (of 5) with a positive difference, and the pre-specified label. frozen-all-gate and frozen+add variants are the same-host tests of the gate in the fully frozen block.

Arm / contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen	80.3	86.4	87.0	
frozen-all	89.5	91.5	88.9	
S4D	68.3	68.3	68.6	
frozen-gate	70.9	70.7	60.0	
frozen+add	71.3	71.3	60.1	
frozen-all-gate	84.1	82.2	56.3	
frozen-all+add	84.9	83.6	65.6	
frozen-all	+21.3 [14.6, 27.9]	+23.3 [20.2, 26.4]	+20.3 [17.9, 22.8]	
– S4D	(5/5)	(5/5)	(5/5)	holds
BM3-frozen	+12.1 [0.9, 23.2]	+18.1 [9.8, 26.4]	+18.4 [12.7, 24.0]	
– S4D	(5/5)	(5/5)	(5/5)	holds
BM3-frozen	+9.4 [−0.5, 19.4]	+15.7 [−0.8, 32.1]	+27.0 [7.0, 46.9]	
– frozen-gate	(5/5)	(4/5)	(4/5)	weakened
frozen-all	+5.4 [−3.8, 14.6]	+9.3 [−0.5, 19.1]	+32.6 [28.9, 36.3]	
– frozen-all-gate	(3/5)	(5/5)	(5/5)	weakened
frozen-all-gate	+15.9 [6.9, 24.8]	+14.0 [2.5, 25.4]	−12.3 [−17.5, −7.0]	
– S4D	(5/5)	(4/5)	(0/5)	descriptive
frozen+add	+0.4 [−0.6, 1.4]	+0.6 [−0.1, 1.2]	+0.1 [−0.4, 0.7]	
– frozen-gate	(4/5)	(5/5)	(3/5)	descriptive
frozen-all+add	+0.8 [−1.5, 3.1]	+1.4 [−1.9, 4.7]	+9.3 [−7.2, 25.8]	
– frozen-all-gate	(2/5)	(2/5)	(5/5)	descriptive

seed-to-seed spread of any arm; the partially frozen block exceeds it by +12.1, +18.1, +18.4 pp.

Gate removal. Removing the gate from the partially frozen block costs +9.4, +15.7, +27.0 pp; the point estimates are large and positive in 13 of 15 seed-level pairs, but only the -6 dB interval excludes zero, so under the pre-specified rule the contrast is *weakened* (it held under the earlier protocol). The same test inside the fully frozen block gives +5.4, +9.3, +32.6 pp, again resolved only at -6 dB. Without its gate, the fully frozen block still exceeds S4D by +15.9 and +14.0 pp at 0 and -2 dB (intervals excluding zero) but falls 12.3 pp below it at -6 dB (interval $[-17.5, -7.0]$, all seeds). At 0 and -2 dB the fully frozen block’s advantage is therefore not carried by the gate; at -6 dB the gate is needed for it. The pre-specified condition for a combined attribution (the fully frozen block’s advantage depends on the gate: gate removal holds and the additive substitution is not rescued) is not met, and we do not make that claim.

Additive substitution. In both hosts the additive arm cannot be distinguished from the gateless arm (+0.4, +0.6, +0.1 pp and +0.8, +1.4, +9.3 pp; all intervals include zero). The point-estimate rescue ratios are small ($\rho = 0.029$, paired-seed interval $[0.011, 0.059]$; in the fully frozen host $\rho = 0.194$, $[-0.002, 0.326]$), but because gate removal is only weakened the pre-specified rule labels the substitution test *not assessable*, and we report it descriptively.

Paderborn, shared bearings. The gate-removal cost holds (+7.6, +6.5, +4.4 pp, intervals excluding zero at all three levels); the partially frozen block’s advantage over S4D is weakened (+5.6, +6.5, +8.3 pp, resolved at -6 dB only).

Table 3: Independent training and evaluation noise, Paderborn shared bearings (KA04, KA16, KI04, KI14; 1500 $\rightarrow$ 900 rpm). Paired differences under final-epoch reporting with 95% t -intervals, seeds positive and pre-specified label. Means: BM3-frozen 81.5 / 83.8 / 85.9, S4D 75.9 / 77.3 / 77.6, frozen-gate 73.8 / 77.3 / 81.5 %.

Contrast	0 dB	-2 dB	-6 dB	label
BM3-frozen – S4D	+5.6	+6.5	+8.3	weakened
	$[-4.4, 15.5]$	$[-2.9, 15.9]$	$[2.7, 13.8]$	
	(4/5)	(4/5)	(5/5)	
BM3-frozen – frozen-gate	+7.6	+6.5	+4.4	holds
	$[2.7, 12.6]$	$[1.7, 11.3]$	$[3.2, 5.6]$	
	(5/5)	(4/5)	(5/5)	

327 4.2. Freezing the input-dependent Δ , A , B , C projections

Table 4: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Sections 4.3 and 4.10). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+15.5
−2 dB	88.3	87.3	67.5	+0.9	+19.8
−6 dB	83.3	83.1	66.7	+0.2	+16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

328 Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs
329 at most 3.7 pp anywhere in the adjudication band on the point estimates
330 under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, −2, −6 dB), while
331 the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D
332 (Table 4, Figure 2). Five seeds bound these quantities loosely: paired 95% in-
333 tervals for $\Delta(\text{BM3-frozen})$ are [−3.8, 11.3], [−4.5, 6.3], [−12.1, 12.6] pp (BM3
334 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ [4.3, 26.8], [12.0, 27.7],
335 [4.1, 28.7] pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robust-
336 ness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the
337 frozen arm collapsing toward S4D. It was written for best-epoch reporting,
338 under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp
339 (re-run: +5.1, +3.5, +3.0) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp.
340 Neither condition holds under either rule, so the pre-specified verdict is that
341 these projections are *not the primary mechanism*. This is a failure to meet
342 a confirmation criterion, not a demonstration that freezing is cheap: the

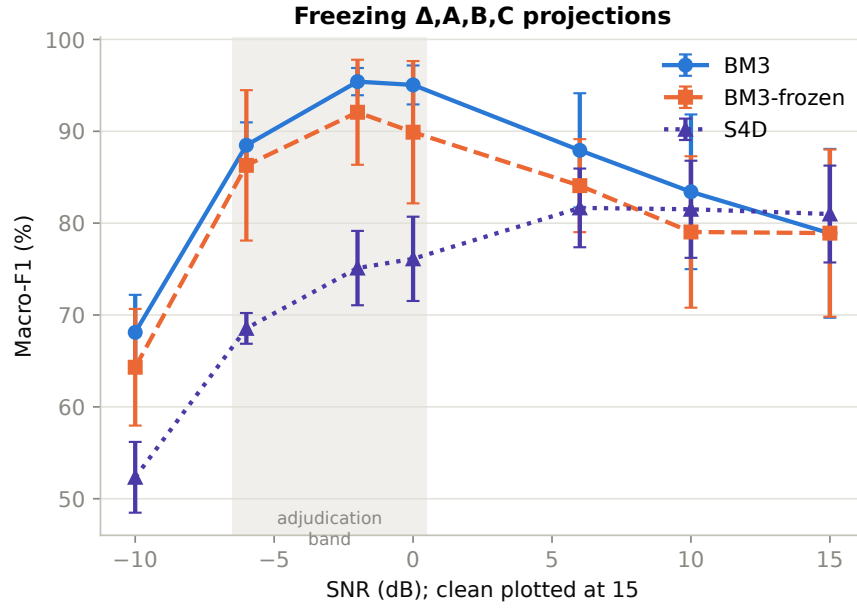


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table 4, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

intervals above are compatible with differences larger than 10 pp, and no equivalence margin was pre-specified or tested. One limit applies: the frozen arm still contains input-dependent trapezoidal weights and rotation angles, so this result concerns Δ, A, B, C only; Section 4.3 removes that limit. On clean data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-style arms—in line with the clean-benchmark comparison of [4], which, like this auxiliary row, involves no added noise. In the noisy regime, the block’s advantage over S4D is not removed by freezing these four terms, so it must be carried by other parts of the block or by the remaining input-dependent scan terms.

4.3. Freezing the remaining input-dependent scan terms

Table 5 reports the three layered arms of Section 3.2 on the adjudication band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training code, the same evaluation fingerprint, with the original BM3 arm re-run in the same grid). The question is whether the advantage that survived freezing Δ, A, B, C is carried by the two input-dependent terms the frozen arm retained.

Table 5: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen–trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen–rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen–all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen–all – S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen – frozen–all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 – frozen–all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

It is not. With the scan’s dynamics and read/write coefficients input-independent, *frozen–all* is +21.9, +24.0 and +21.2 pp above S4D at 0, −2

and -6 dB under final-epoch reporting, positive in 5/5 seeds at every level and with intervals excluding zero—if anything a larger margin than the partially frozen arm’s $+15.5$, $+19.8$, $+16.4$ pp (Table 13). None of the three freezing steps costs accuracy: the paired differences BM3-frozen – frozen–trap, – frozen–rot and – frozen–all are negative at all three levels (between -1.8 and -6.3 pp under final-epoch reporting), i.e. the more heavily frozen arm is numerically ahead, with every interval including zero. The direction is not an artefact of capacity: *frozen–all* has 103,842 parameters against 112,410 for BM3-frozen and 177,938 for BM3. At the deepest level *frozen–all* also exceeds the full BM3 block by 4.5 pp ($[-7.2, -1.9]$ for BM3 – frozen–all, 0/5 seeds in BM3’s favour); we report this but do not build on it, since five seeds and one speed pair cannot separate a genuine advantage of a time-invariant scan from a variance effect.

Two readings of the whole step are therefore available, and the data do not separate them: input-dependence of the scan contributes nothing measurable here, or it contributes something that the learned constants recover under this training budget. Either way the conclusion for the attribution chain is the same and now rests on a complete intervention rather than a partial one: **the Mamba-style block’s deep-noise advantage over a time-invariant S4D does not require any input-dependent term in its scan**. What remains to carry it is the rest of the block—the convolutional stem, the heavy-tailed A parameterisation, the normalisations and the output gate—which is what Sections 4.4–4.5 examine. The re-run also reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build, Section 4.10).

4.4. *Removing the gate is the only ablation with a large, one-signed effect*

“Necessary” is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block’s advantage. The gate result is large, one-signed and grows with noise depth: under final-epoch reporting removal costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals $[2.9, 15.3]$, $[-0.8, 32.0]$, $[5.8, 39.4]$; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at -2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ($[6.5, 19.6]$, $[3.8, 38.5]$, $[8.5, 38.6]$; 5/5 seeds at every level), so under

Table 6: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Section 4.10); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; intervals include 0

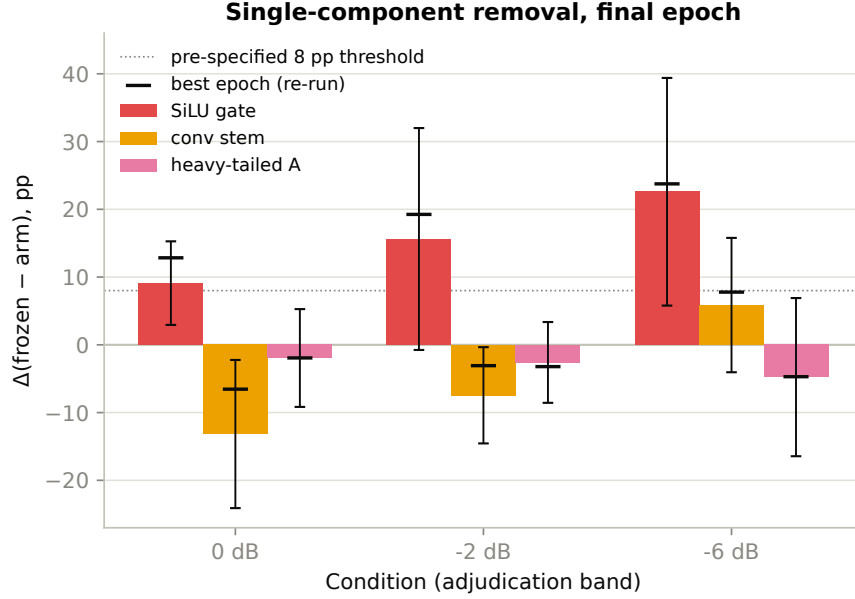


Figure 3: Component-removal Δ (percentage points, frozen $-$ arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold.

399 the earlier noise protocol the verdict is *primary* under both rules. Under
 400 independent training and evaluation noise the same contrast is resolved only
 401 at -6 dB (Section 4.1), so the operational sense of “necessary” holds there
 402 at the deepest level only. The gateless arm gives up most of the Mamba
 403 advantage but is not resolved against plain S4D: frozen+gate – S4D is $+6.4$,
 404 $+4.2$, -6.2 pp, with every interval including zero. We therefore describe it
 405 as roughly at the level of S4D, below it only on the point estimate at -6 dB;
 406 this suggests, but does not establish, that components interact rather than
 407 add. The other two components tell regime stories rather than robustness
 408 stories. Removing the convolutional stem is beneficial at 0 and -2 dB (frozen
 409 – frozen+conv = -13.2 [-24.1 , -2.2] and -7.5 [-14.6 , -0.3], with the stem-
 410 less arm ahead in every seed) and costly at -6 dB ($+5.9$, [-4.0 , 15.8], 3/5
 411 seeds); under best-epoch reporting the pattern was the same (-6.6 , -2.7 ,
 412 $+7.9$ pp). On clean cross-speed data (auxiliary, best epoch, original grid)
 413 the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing
 414 from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pat-
 415 terns mistransferring across conditions, or with optimisation friction under
 416 the matched budget; we report both readings. Replacing the heavy-tailed
 417 state parameterisation by an S4D-style one is numerically beneficial at all
 418 three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under
 419 best-epoch reporting), but every paired interval includes zero, so we draw no
 420 directional conclusion from it.

421 4.5. Adding a narrow gate branch to S4D

422 The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-}$
 423 $\text{S4D})/(\text{frozen} - \text{S4D})$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level
 424 0.249, 0.341, -0.003 ; seed-resampled 95% interval $[0.09, 0.31]$); under the
 425 best-epoch rule for which it was specified, $R = 0.174$ ($[0.04, 0.29]$). The
 426 pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recov-
 427 ery) and $R < 0.3$ (insufficient); both values fall in the last, although the
 428 final-epoch interval reaches the 0.3 boundary. The per-level paired gains of
 429 S4D+gate over S4D under final-epoch reporting are $+3.9$ ($[-0.8, 8.6]$), $+6.8$
 430 ($[2.4, 11.1]$), -0.1 ($[-4.0, 3.9]$) pp. At the deepest adjudicated level the graft
 431 recovers nothing. Removal and addition are not mirror-image interventions.
 432 In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan
 433 output at the inner width of 128 before the output projection; removing it
 434 deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$
 435 linear branch (with bias) computed from the layer input and multiplying the

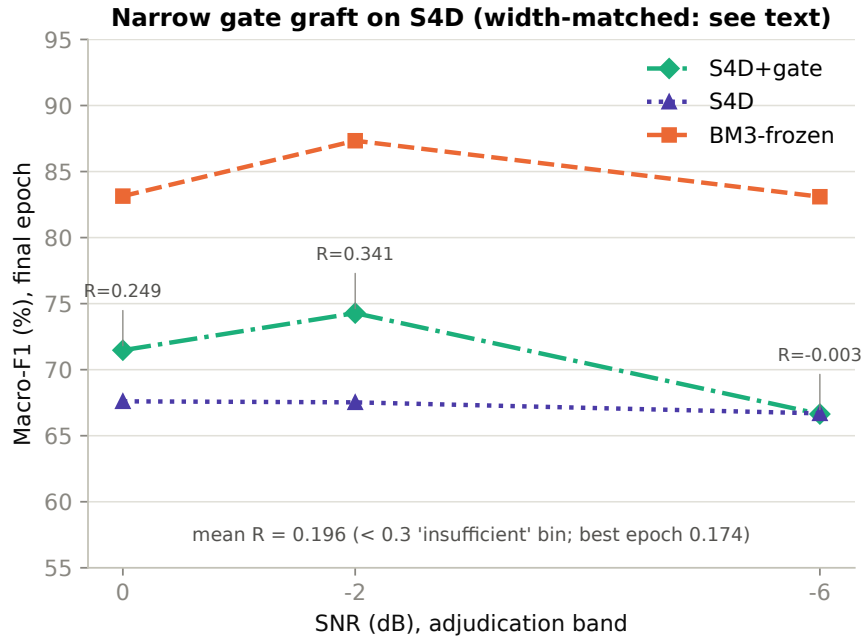


Figure 4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (S4D+gate - S4D) / (frozen - S4D)$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 8.

Table 7: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Section 4.10). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Section 4.4 (independent runs of the same arms differ by 0.1–0.7 pp). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 9–23 pp in the frozen block, whereas this graft changes S4D by between −0.1 and +6.8 pp (final epoch). What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Section 4.6 tests reading (ii) with a width-matched graft and an equal-capacity non-multiplicative control; a tuning or longer-budget comparison selected on a source-domain validation set was not run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp at 0, −2 and −6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4 [−1.4, 30.3] and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more

on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.6. A width-matched graft and an additive control

The graft of Section 4.5 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host’s 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate (wm)* combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

Table 8: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of Section 4.10. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB	−2 dB	−6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				−0.112	[−0.26, 0.02]

478 The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159,
 479 0.298, 0.431). Its interval, $[0.16, 0.52]$ under the pre-specified bootstrap
 480 $([0.14, 0.72]$ under paired-seed resampling), crosses the 0.3 boundary, so the
 481 pre-specified verdict is INCONCLUSIVE between “insufficient” and “par-
 482 tial”, and we report it at the less favourable tier: a graft built in the donor’s
 483 shape recovers part of the gap. The recovery is not gate-specific. The addi-
 484 tive branch, with the same parameters, recovers at least as much ($R = 0.408$),
 485 $\Delta R = -0.112$ $[-0.26, 0.02]$ (paired-seed $[-0.20, 0.06]$), and at 0 dB the gat-
 486 ed arm is below the additive one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With
 487 frozen—all in the denominator the gated arm’s ratio is 0.231 $[0.13, 0.33]$; un-
 488 der best-epoch reporting the two ratios are 0.459 and 0.483. What a graft
 489 recovers on this transition therefore follows the added structure—a wider
 490 channel axis and input/output projections—rather than the multiplicative
 491 interaction. Reading (ii) of Section 4.5 is partly confirmed, in that the nar-
 492 row graft understated what added structure can recover; but in the S4D host
 493 the gate did not contribute measurably beyond an equal-capacity additive
 494 branch, whereas removing it from the frozen Mamba-3 block costs 9–23 pp
 495 under the same rule (Table 13).

496 4.7. Replacing the gate in its native host

497 The previous subsection changes the host and keeps the operator. A pre-
 498 specified grid (15 cells, original transition) does the converse: it keeps the host
 499 and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen
 500 layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$;
 501 modules, parameter count (112,410) and initial weights are bitwise those of
 502 BM3-frozen, and a construction check confirms that the donor operator is
 503 $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified
 504 statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$,
 505 mean over levels, final epoch, with a seed-resampled interval; *not rescued*
 506 requires $\rho < 0.3$ with an upper bound below 0.5.

507 The additive branch does not measurably rescue the loss. *frozen+add*
 508 cannot be distinguished from the gateless arm at any level (+0.6, −0.2,
 509 +0.4 pp, every interval including zero), and the multiplicative arm exceeds
 510 it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070,
 511 −0.015, 0.016), with a 95% interval of $[-1.159, 0.486]$ under the pre-specified
 512 bootstrap, which resampled each arm and level independently; resampling
 513 seeds jointly across arms and levels, as the paired design warrants, gives
 514 $[-0.043, 0.136]$. Under best-epoch reporting $\rho = 0.005$. By the pre-specified

Table 9: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN $\rightarrow$ 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5		+15.8		+22.2	
	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6		−0.2		+0.4	
	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
BM3-frozen − frozen+add	+18.1		+22.3		+23.4	
	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
frozen+add − frozen−gate	+0.3		−0.3		+0.6	
	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

rule the verdict is *not rescued*: the point-estimate rescue is small and below the pre-specified criterion, which is not a demonstration that the two arms are equivalent.

A further pre-specified grid (15 cells) repeats the substitution on the reverse transition (Section 4.8), with BM3-frozen and frozen+gate taken from the reverse grid. The outcome is the same (Table 9, lower block): frozen+add cannot be distinguished from the gateless arm at any level (+0.3, -0.3, +0.6 pp, every interval including zero), and the multiplicative arm exceeds it by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, -0.014, 0.023), with a 95% interval of $[-0.943, 0.362]$ under the pre-specified bootstrap and $[0.002, 0.018]$ under paired-seed resampling; under best-epoch reporting $\rho = -0.005$. The pre-specified verdict is again *not rescued*.

Together with Section 4.6 this gives the two halves of a symmetric test. In the S4D host the gate did not recover more than an additive branch of the same size on the original transition and recovered somewhat more on the reverse one; in the Mamba-3 host an additive branch of the same size could not be distinguished from removing the gate, on both transitions. Under the earlier noise protocol this pattern is specific to the multiplicative operator in its native host; Section 4.1 shows that the size of the gate’s contribution there depends on noise depth.

4.8. The reverse transition

A pre-specified grid repeats the chain’s central contrasts on the reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen-all, BM3-frozen, frozen+gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). A second pre-specified grid (60 cells) re-ran the first four arms with independent training and evaluation noise; their values below are from that re-run, which moved the arm means by -1.8 to +1.7 pp relative to the earlier protocol. This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8% and uniform random guessing 41.5%, and S4D sits close to those levels (Table 10).

The first two steps replicate under independent noise, all three pre-specified contrasts with the label *holds*. The fully frozen block exceeds S4D by +27.5, +26.7, +21.1 pp at 0, -2 and -6 dB and the partially frozen block by +25.5, +27.5, +20.0 pp, with intervals excluding zero and 5/5 seeds at every level. Removing the gate from BM3-frozen costs 19.3, 22.7, 22.0 pp, above the

Table 10: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch / final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference. Rows for frozen—all, BM3-frozen, frozen—gate and S4D and the first four contrasts use independent training and evaluation noise (60-cell re-run); S4D+gate (wm) and its contrast use the earlier noise protocol (training and evaluation noise shared by index), against the earlier-protocol S4D.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen—all	88.6 / 77.4	88.1 / 77.6	81.1 / 73.3
BM3-frozen	85.5 / 75.4	86.7 / 78.4	78.5 / 72.2
frozen—gate	64.0 / 56.1	62.5 / 55.7	55.1 / 50.1
S4D	61.2 / 49.9	60.2 / 50.9	55.7 / 52.2
S4D+gate (wm) [†]	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
	+27.5	+26.7	+21.1
frozen—all − S4D	[21.3, 33.8] (5/5)	[19.5, 33.9] (5/5)	[16.2, 26.0] (5/5)
	+25.5	+27.5	+20.0
BM3-frozen − S4D	[13.9, 37.1] (5/5)	[18.2, 36.9] (5/5)	[8.0, 32.0] (5/5)
	+19.3	+22.7	+22.0
BM3-frozen − frozen—gate	[2.9, 35.7] (5/5)	[4.1, 41.4] (5/5)	[3.2, 40.9] (5/5)
	+6.2	+4.8	−2.0
frozen—gate − S4D	[−15.6, 28.0] (2/5)	[−17.5, 27.1] (2/5)	[−19.6, 15.5] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) − S4D [†]	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

[†]Earlier noise protocol.

8 pp threshold at all three levels, with intervals excluding zero at all three levels and 5/5 seeds, and the gateless arm cannot be distinguished from S4D (intervals include zero). In this direction, therefore, gate removal is resolved at every level, whereas on the original transition under the same noise protocol it is resolved only at -6 dB (Section 4.1). On the third step, which the pre-specification treats as descriptive here and which was not re-run (earlier noise protocol), the width-matched graft recovers most of the gap, $R = 0.911$ (paired-seed interval $[0.82, 1.07]$) (0.863 with frozen—all in the denominator). A further pre-specified grid (15 cells) adds the same-size additive control in this direction: it reaches 67.8, 72.1, 68.4% (final epoch) and recovers $R = 0.758$ $[0.64, 0.90]$. The gate arm exceeds it by $+5.1$, $+1.7$, $+3.8$ pp (only the -6 dB interval excludes zero), and the difference in recovery is $\Delta R = 0.153$ (paired-seed interval $[0.01, 0.34]$); under best-epoch reporting it shrinks to 0.037 $[-0.1, 0.17]$. By the pre-specified descriptive labels this is *unresolved*: in this direction the gate carries a small increment over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

4.9. External validity: two settings on a second dataset

A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the writing consequences committed before the first cell) repeats the three central contrasts on the Paderborn KAt data [16], using real-damage bearings only. Models train on two outer-race (KA04, KA16) and two inner-race bearings (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15, KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged; the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms as in XJTU-SY, and everything else (noise, training, arms, budget) is as in the main grids. The pre-specified smoke check (two arms, one extra seed) passed its three criteria, the learnability criterion by 0.3 pp. A second pre-specified grid (45 cells, three arms) then used the same four bearings for training and evaluation, at the same two speeds, from different recordings. This changes the evaluation bearings, their damage profiles, the number of

windows and the class balance together, so the two settings are two protocols rather than a single-factor manipulation.

Table 11: Paderborn, real-damage bearings, 1500 rpm $\rightarrow$ 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t -interval and seeds (of 5) with a positive difference. “-”: arm not run in that setting. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm / contrast	Disjoint bearings			Shared bearings		
	0 dB	-2 dB	-6 dB	0 dB	-2 dB	-6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen-all	54.2	51.3	47.8	-	-	-
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen-gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	-	-	-
	-3.8	-3.6	-3.4	+8.4	+8.5	+8.7
BM3-frozen	[-11.9, 4.3]	[-12.6, 5.4]	[-8.9, 2.1]	[-1.3, 18.1]	[0.2, 16.8]	[4.3, 13.0]
- S4D	(1/5)	(2/5)	(1/5)	(4/5)	(5/5)	(5/5)
	-5.0	-4.6	-3.3	+9.0	+7.7	+5.3
BM3-frozen	[-8.8, -1.3]	[-9.2, 0.1]	[-7.1, 0.4]	[5.2, 12.8]	[5.0, 10.4]	[2.4, 8.2]
- frozen-gate	(0/5)	(0/5)	(0/5)	(5/5)	(5/5)	(5/5)
	-1.4	-2.5	-3.3			
frozen-all	[-4.0, 1.2]	[-5.3, 0.2]	[-5.8, -0.8]			
- S4D	(1/5)	(1/5)	(0/5)	-	-	-
	0.0	0.0	-0.1			
frozen+add	[-0.9, 0.8]	[-0.4, 0.3]	[-0.4, 0.2]			
- frozen-gate	(2/5)	(3/5)	(1/5)	-	-	-

None of the pre-specified contrasts replicates (Table 11). The fully input-independent block does not exceed S4D (frozen-all - S4D: -1.4, -2.5, -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor does BM3-frozen (-3.8, -3.6, -3.4 pp). Removing the gate does not cost accuracy: BM3-frozen - frozen-gate is -5.0, -4.6, -3.3 pp, below zero in every seed at every level. With no gate effect to rescue, the native-substitution test is not assessable; the additive and gateless arms again cannot be distinguished (0.0, 0.0, -0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze contrasts and for gate removal, and *not assessable* for the substitution test.

Two observations bound how this result can be read, and neither changes the labels. First, the testbed discriminates poorly: every arm ends between

47.7 and 56.9% macro-F1 at every level (majority-class predictor 37.5%, a balanced random predictor about 50%), all arms fit the training condition (the gated arms reach the lowest training loss), and none transfers far above the level reached after the first epoch. Second, where differences are resolved they run against the Mamba-3 arms, which reach lower training loss and transfer slightly worse. In the disjoint setting, therefore, the block’s advantage under matched noise and drift is not observed; we do not know whether this reflects a property of the PU transition (speed reduction to 900 rpm with disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a limit of the XJTU-SY result itself, which the shared-bearing setting below separates.

Shared bearings. With bearing identity shared between training and evaluation, the same condition pair becomes transferable: every arm ends between 75.5 and 86.6% at every level, and the pre-specified transferability gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two of three levels for at least one arm) is passed by all three arms at all three levels (smallest lower bound 71.3%). Within this setting both effects replicate under the pre-specified rule (Table 11, right). BM3-frozen exceeds S4D by +8.4, +8.5, +8.7 pp at 0, -2 , -6 dB, with intervals excluding zero at -2 and -6 dB; removing the gate costs +9.0, +7.7, +5.3 pp, with intervals excluding zero at all three levels and in every seed; no level runs in the opposite direction. The gateless arm returns to the level of S4D (frozen–gate – S4D: -0.6 , $+0.8$, $+3.4$ pp, all intervals including zero), as on XJTU-SY. Under best-epoch reporting the freeze contrast is resolved at all three levels (+6.9, +10.8, +13.7 pp) and gate removal only at -6 dB (+1.5, +3.7, +6.4 pp). The cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level only; we compare directions, not magnitudes, across datasets.

Read together, the two protocols give different results: with shared bearings every arm, S4D included, ends 18.6–38.8 pp higher than with disjoint bearings (independent seeds; per-arm Welch intervals all exclude zero), and the block’s advantage and the cost of gate removal reappear. Because the evaluation bearings, class balance and sample size change together, this does not isolate bearing overlap as the cause. With two bearings per class in the shared setting, a bearing’s identity also determines its label, so a model can succeed partly by recognising bearing-specific signatures. XJTU-SY differs from both Paderborn settings: its training and evaluation bearings are disjoint in each direction, and the block transfers there. The shared-setting contrasts were re-run under independent noise (Section 4.1); *frozen–all* was

not run on Paderborn.

4.10. Sensitivity to the reporting rule

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen-gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Section 4.3), and the capacity control and the two other single-component removals, S4D-wide, frozen-conv and frozen-A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp across the eight re-run arms (the largest deviation is in the high-variance frozen-gate arm at -2 dB).

Table 12: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen-conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen-A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Best-epoch values exceed final-epoch values by 0.6–10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen

659 -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0, -2 , -6 dB), and the
660 contrasts that carry the argument are correspondingly stable (Table 13).

Table 13: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Contrast	0 dB	-2 dB	-6 dB
	+15.5	+19.8	+16.4
BM3-frozen $-$ S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen $-$ frozen-gate	[2.9, 15.3] (5/5)	$[-0.8, 32.0]$ (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	-6.2
frozen-gate $-$ S4D	$[-2.9, 15.8]$ (4/5)	$[-14.5, 22.9]$ (3/5)	$[-23.5, 11.2]$ (1/5)
	+3.9	+6.8	-0.1
S4D+gate $-$ S4D	$[-0.8, 8.6]$ (4/5)	[2.4, 11.1] (5/5)	$[-4.0, 3.9]$ (2/5)

661 *Selectivity.* The frozen block’s advantage over S4D is +15.5, +19.8 and
662 +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, Section 4.3),
663 versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the re-run.
664 It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero
665 at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to
666 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified
667 threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp,
668 4/5 seeds) because of the large seed variance of the gateless arm (standard de-
669 viation 14.6 pp at -2 dB). *Gateless versus S4D.* Under final-epoch reporting
670 the gateless arm is not below S4D at 0 and -2 dB (point estimates +6.4 and
671 +4.2 pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero.
672 The statement that gate removal “falls below S4D” is therefore supported
673 only as a point-estimate observation at the deepest level, and we phrase it as
674 “to or below” elsewhere. *Graft.* The recovery ratio is 0.249, 0.341 and -0.003
675 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the
676 pre-specified “insufficient” bin on the point estimate. The seed-resampled
677 95% interval is [0.09, 0.31] and so reaches the 0.3 boundary: with five seeds
678 the data support “far below confirmation (0.7) and probably below 0.3”, not
679 a precise value of R . The graft gain over S4D is resolved at -2 dB only
680 (+6.8 pp, [2.4, 11.1]) and is indistinguishable from zero at -6 dB.

681 *Remaining arms.* By the rule pre-specified for the 45-cell grid, no arm
682 changes sign between the two rules at any level where both values exceed

1 pp, so all main-table entries are reported under the final-epoch rule with unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-wide by +15.9, +20.8, +16.0 pp (intervals excluding zero at all three levels), and S4D-wide – S4D is –0.4, –1.0, +0.4 pp (intervals including zero). For the stem the final-epoch rule sharpens the regime trade-off (intervals at 0 and –2 dB now exclude zero) without reversing it.

Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, last-epoch values are also not a fresh confirmatory test. The reporting rule therefore changes absolute levels and the strength of the gateless-versus-S4D statement, but not the ordering BM3-frozen > {S4D+gate, frozen–gate, S4D}, the size of the frozen block’s advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

4.11. The interaction effect concentrates on the hard class

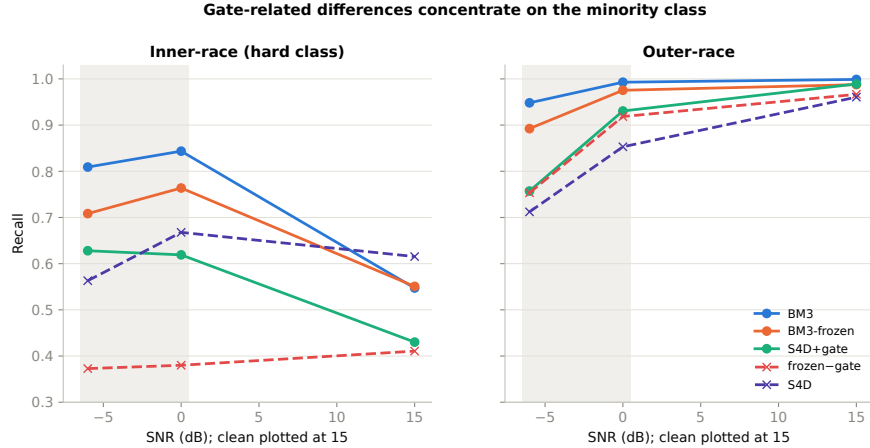


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: $5 \text{ arms} \times \{\text{clean}, 0, -6 \text{ dB}\} \times 5 \text{ seeds}$, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at –6 dB exceeds its clean value (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its

703 clean value (frozen—gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone,
 704 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
 705 gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from
 706 predictions pooled over seeds), and the graft’s noise-time gains also concen-
 707 trate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages
 708 the two classes, the arm-level robustness gaps of Sections 4.2–4.5 are carried
 709 substantially by the hard class. The gate-related differences are therefore
 710 class-concentrated rather than uniform, and fall on the class with the lowest
 711 recall. We did not measure classification margins, so we do not describe IR as
 712 a “low-margin” class. Because this grid is separate from the main 50-epoch
 713 grid and the IR class comes from a single training bearing (Section 3.1), these
 714 results are descriptive and do not decompose the main-grid effects.

715 5. Discussion

716 **What can be credited?** The ablation chain supports a narrow, con-
 717 ditional statement: within this Mamba-3 block, on this testbed and with
 718 noise-aware training, (i) making the scan fully input-independent—first the
 719 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
 720 well—did not remove the block’s deep-noise advantage over a time-invariant
 721 S4D, and at the deepest level the fully frozen arm matched or exceeded the
 722 original block with fewer parameters than BM3 or BM3-frozen (though more
 723 than S4D), (ii) removing the output gate removed most of that advantage—
 724 (i) and (ii) holding on the reverse transition as well, where under independent
 725 noise gate removal is resolved at every level—and (iii) adding the gate to S4D
 726 recovered part of the advantage only together with added width and projec-
 727 tions, and an additive branch of the same size recovered as much, while (iv)
 728 in the Mamba-3 block an additive branch of the same size could not be dis-
 729 tinguished from removing the gate, on both transitions. Under independent
 730 training and evaluation noise (Section 4.1), (i) holds, (ii) is resolved only
 731 at -6 dB, and in the fully frozen block the gate carries the advantage at
 732 -6 dB but not at 0 and -2 dB. These observations are consistent with a
 733 gate benefit that is specific to the multiplicative operator in its native host
 734 and grows with noise depth. They do not identify which other component
 735 the gate depends on. Nor do they separate “input-dependence contributes
 736 nothing here” from “what it contributes is recovered by learned constants
 737 under this budget”; the two readings are indistinguishable in our data and
 738 lead to the same attribution. Representability results for selectivity [8] are

not contradicted by a single ablation on one testbed. The S4D-superiority finding [4] was obtained on clean classification benchmarks, without matched noise or a shift of operating condition, so it does not speak to the regime studied here and we do not read it as covering it. Two points of contact are nevertheless consistent with it: our clean-data parity between S4D and the Mamba-style arms, and our finding that input-dependent selectivity is not needed for the block’s advantage. What this paper adds is specific to the matched-noise, cross-condition regime: there S4D trails the frozen Mamba-3 block by 12–28 pp under final-epoch reporting and independent training and evaluation noise (partial and full freeze, both transitions), and the gap does not require input-dependent scan coefficients; at the deepest level it depends on the multiplicative output gate. This is a statement about a different regime, not a contradiction of the clean-benchmark result.

Implications for backbone selection and for transplant practice.

For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest that the block’s benefit should be judged as a composition, and that a gate added to a different host should not be expected to reproduce it: in our S4D host a gate built in the donor’s shape recovered no more than an additive branch of the same size on the original transition (Section 4.6) and only slightly more on the reverse one (Section 4.8), and neither design was tuned. A hypothesis for future work follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style A) may match or exceed the original on this testbed; we leave it untested, because post-hoc recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Section 4.11), and matched-noise training lifts that class *only when the gate is present*. One hypothesis, which we did not test, is that the gate is the component through which matched-noise training improves hard-class recall, i.e., that it exploits noise-aware training rather than acting as a test-time denoiser; we measured neither classification margins nor gate activations, and the reading is in any case bounded by the matched-SNR scope (Section 6). Second, the multiplicative form admits a per-sample suppression interpretation, but no gate statistics were recorded, so this remains a hypothesis rather than an observation. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other compo-

777 nents being calibrated to a gated regime (e.g., the stem may pass features
 778 that are only safe when the gate can veto them); this is the weakest of the
 779 three observations, since the gateless-S4D gap is not resolved with five seeds
 780 (Section 4.10). All are testable in future work via pairwise interaction grids
 781 and mismatched-noise protocols.

782 6. Boundaries and Limitations

783 We state the scope plainly.

- 784 1. **Testbed scope.** All claims are bounded to bearing vibration from
 785 one pair of operating conditions of one dataset (37.5 Hz/11 kN and
 786 40 Hz/10 kN; speed and load both change), binary classification, G-
 787 PU execution (no edge/real-time claims). Within each direction the
 788 training and evaluation bearings are disjoint; the pair is run in both
 789 directions, but the reverse direction uses the same eight bearings with
 790 their roles swapped and has a single IR bearing in its evaluation set
 791 (the forward direction a single IR bearing in its training set), so it
 792 tests direction sensitivity rather than replicating on new bearings; the
 793 third condition of the dataset has no IR bearing. The five seeds do
 794 not resample bearings (Section 3.1), so the intervals carry no bearing-
 795 level uncertainty and class-level results may reflect individual bearings.
 796 Further independent datasets and multi-class settings are future work.
- 797 2. **Attribution level.** Credit assigned to “the rest of the composition”
 798 is by elimination within the enumerated component set; pairwise in-
 799 teraction terms are not decomposed, and the host-by-gate contrast of
 800 Section 4.5 is a four-arm estimate that cannot be assigned to a specific
 801 component.
- 802 3. **The transplant conclusion is budget-conditional.** All arms share
 803 a matched 50-epoch budget; the grafted gate trains normally yet re-
 804 covers little, and convergence is not robustness (Section 4.5)—but we
 805 cannot exclude that a longer or graft-tuned budget would recover more.
 806 Our statement is therefore limited to *this gate branch, in this host, un-*
 807 *der a matched training budget.*
- 808 4. **Matched-SNR scope.** Training and evaluation share the same in-
 809 jected SNR throughout (Section 3.1); “robustness” in this paper is
 810 matched-noise robustness under noise-aware training, not clean-train/noisy-
 811 test generalisation. The gate’s benefit may operate by exploiting matched-
 812 noise training (consistent with Section 4.11) rather than by test-time

- 813 denoising; attribution under mismatched-noise protocols is untested
814 and may differ.
- 815 5. **Graft dual reading.** Interaction dependence vs. transplant friction
816 (Section 4.5), the latter now subsumed under the budget condition of
817 item 3.
- 818 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
819 TCN baseline slightly overtakes the Mamba-style block, and -10 dB
820 is near-chance for the time-invariant baseline and excluded from ad-
821 judication; the demonstrated advantage regime is matched, Gaussian-
822 dominated noise.
- 823 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
824 mistransfer and optimisation-difficulty readings, reported in place (Sec-
825 tion 4.4).
- 826 8. **Reporting rule.** Reported values are final-epoch macro-F1 under a
827 fixed 50-epoch budget. The original grids’ best-epoch rule, for which
828 the thresholds of the early grids were pre-specified, is reported as a
829 sensitivity analysis and is optimistic by 0.6–10.6 pp (Section 4.10). Ev-
830 ery arm of the chain was re-run on the three adjudication levels; the
831 $+10/+6/clean/-10$ dB levels and the per-class grid were not re-run
832 and are auxiliary (best epoch). No source-domain validation rule was
833 available, and because the final epoch of the same evaluation condi-
834 tion has been inspected repeatedly, final-epoch values are not a fresh
835 confirmatory test.
- 836 9. **Scope of the freezing intervention.** The chain’s first step freezes
837 the input-dependent parts of Δ , A , B , C only; the fully input-independent
838 scan is a separate grid (Section 4.3) on the three adjudication levels,
839 which was specified after the earlier grids had been seen. No equivalence
840 margin was pre-specified for the cost of freezing, so “no measurable
841 cost” means the point estimates favour the frozen arms with intervals
842 including zero, not a demonstrated equivalence. Freezing is implement-
843 ed inside this Mamba-3 layer; a conclusion about input-dependence in
844 general would need other blocks and tasks.
- 845 10. **Gate interventions.** Under independent noise, gate removal is re-
846 solved only at -6 dB in both the partially and the fully frozen host,
847 and without its gate the fully frozen block still exceeds S4D at 0 and
848 -2 dB, so the combined statement that the fully frozen block’s advan-
849 tage is carried by the gate is not supported. The additive substitu-
850 tion could not be distinguished from gate removal in either host; with

- gate removal only weakened, the pre-specified rule leaves it not assessable under independent noise (*not rescued* under the earlier protocol, in both directions). In S4D the width-matched arms are larger than the donor block, and the gate’s increment over the additive branch is direction-dependent (paired-seed interval for the reverse-direction difference [0.013, 0.338]).
11. **Noise protocol.** All grids before Section 4.1 used training and evaluation noise shared by index; their arm means differ from the independent-noise re-run by at most 4 pp where both exist, but grids that were not re-run (component ablations of the stem and A , grafts and additive substitutions on the reverse transition, per-class grid) carry that protocol.
 12. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; for all but the last four follow-up grids the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.
 13. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.
 14. **Second dataset.** On Paderborn, the gate-removal cost replicates when training and evaluation share bearings (also under independent noise), the partially frozen block’s advantage only at -6 dB, and nothing replicates on bearings disjoint from the training set, where every examined architecture, S4D included, transferred poorly (47.7–56.9% against a balanced-chance level of about 50%). The two Paderborn protocols differ in more than bearing overlap, so they do not identify why transfer fails; *frozen—all* was not tested on Paderborn.
 15. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

888 7. Conclusion

889 In this Mamba-3 block, on one controlled shift $\times$ noise testbed with
890 matched-noise training, making the scan’s dynamics and read/write coef-
891 ficients input-independent did not remove the block’s advantage over a time-
892 invariant S4D; under independent training and evaluation noise the fully
893 frozen block leads by +21.3, +23.3, +20.3 pp, and on the reverse transition
894 by +27.5, +26.7, +21.1 pp. The multiplicative output gate is the ablated
895 component whose removal costs most, but under independent noise this cost
896 is resolved only at -6 dB on the original transition (at every level on the
897 reverse one), and without its gate the fully frozen block still exceeds S4D at
898 milder noise: the gate’s contribution grows with noise depth rather than car-
899 rying the advantage throughout. An additive branch of identical size, placed
900 where the gate was, could not be distinguished from removing it; added to
901 S4D, a gate recovers part of the gap, not more than an additive branch of the
902 same size on one direction and somewhat more on the other. On Paderborn,
903 the gate-removal cost replicates when bearings are shared between training
904 and evaluation and nothing replicates on disjoint bearings, where every ar-
905 chitecture transfers poorly. The evidence rests on one block, one shift pair
906 per dataset, five seeds and final-epoch reporting under a fixed budget. We
907 offer the three-step chain—freeze, remove, replace—as a protocol for turning
908 architecture comparisons into component-level tests, with two cautions from
909 this study: interventions must be matched in width and capacity, and the
910 noise streams of training and evaluation must be independent.

911 CRediT authorship contribution statement

912 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
913 analysis, Investigation, Data curation, Writing – original draft, Writing –
914 review & editing, Visualization, Supervision, Project administration.

915 Declaration of competing interest

916 The authors declare that they have no known competing financial inter-
917 ests or personal relationships that could have appeared to influence the work
918 reported in this paper.

919 **Declaration of Generative AI and AI-assisted technologies in the**
920 **writing process**

921 During the preparation of this work the authors used LLM-based agents
922 to execute and cross-check pre-specified analysis and adjudication passes as
923 described in Appendix C, and to assist with manuscript drafting and figure
924 generation. After using these tools, the authors reviewed and edited the con-
925 tent as needed and take full responsibility for the content of the published
926 article. All numeric thresholds, adjudication rulings, and scientific interpre-
927 tations reported in this paper were fixed and made by the human authors,
928 as detailed in Appendix C.

929 **Data availability**

930 The code, pre-specification artifacts, and adjudication trail for this study
931 are available at <https://github.com/jeff-1980/bm3-jizi>. The reposi-
932 tory includes all follow-up grids (645 cells with per-epoch curves, including the
933 210 independent-noise cells of Sections 4.1 and 4.8), their pre-specifications,
934 drivers, construction checks and decision memos and the harness’s own im-
935 ported modules, so the training entry point runs from a fresh checkout.

936 **Appendix A. Fairness protocol**

937 *Appendix A.1. Evaluation-set fingerprint*

938 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
939 256 hash of the raw held-out test-label byte array, computed once per run
940 and asserted identical across every arm/condition/seed cell within a grid.
941 It fingerprints *that the same held-out label vector was scored every time*, not
942 model weights or predictions. It is also the split on which the best-epoch rule
943 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
944 five grids is:

945 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

946 This value is recomputed independently at adjudication time (not merely
947 copied from the run log) and cross-checked row-by-row against every grid’s
948 fairness report before any comparison across arms is treated as valid.

949 *Appendix A.2. Per-arm parameter counts*

950 Table A.14 lists every arm’s parameter count, each obtained directly via
 951 `sum(p.numel() for p in model.parameters())` on the instantiated model
 952 (not estimated), cross-verified against an independently maintained archi-
 953 tecture map.

Table A.14: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

954 *Appendix A.3. Five-batch pre-specification timeline*

955 Table A.15 lists, for each of the five pre-specified grids, the artifact that
 956 fixed its adjudication thresholds, that artifact’s creation/modification date,
 957 and the corresponding grid’s run-directory start time. In every case the
 958 prereg artifact predates the grid it governs.

959 Two honest caveats. First, the baseline batch predates the orchestration
 960 infrastructure used for the other four batches by five days; its prereg trail
 961 rests on the task card’s creation timestamp alone, not on the same automat-
 962 ed executor/reviewer machinery documented in [Appendix C](#), and we state
 963 this rather than implying a uniform process across all five batches. Second,
 964 for the component-ablation grid, two additional JSON files bearing “pre-
 965 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
 966 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
 967 time; both are explicitly marked void in the corresponding decision artifact’s
 968 `void_preregs` field, and neither was used to adjudicate any result in this pa-
 969 per. They are the on-disk residue of the first post-hoc criterion substitution
 970 described in [Appendix C](#), retained here (rather than deleted) as part of the
 971 released adjudication trail.

Table A.15: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ab- lation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

972 Appendix B. Confusion matrices and a repeat-run check

973 Appendix B.1. Full confusion-matrix set

974 Table B.16 gives the complete confusion-matrix set underlying the per-
975 class analysis of Section 4.11: 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
976 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
977 counts were dumped per-sample and re-verified against the reported recal-
978 l/precision/F1 metrics cell-by-cell before use.

Table B.16: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
frozen—gate	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
S4D	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

979 Appendix B.2. Repeat-run check

980 Because two independent process runs of an identical configuration are
981 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
982 ductions in the selective-scan and normalisation ops), we measured a process-
983 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
984 clean, seed 0, 2 epochs) was run twice in separate processes with identical
985 evaluation fingerprints and identical parameter counts confirmed. The two
986 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

987 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
988 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
989 run, and therefore neither an estimate nor a bound of process variability for
990 full-length runs. It is an order of magnitude below the seed-level variabil-
991 ity observed within the deep-noise band across the curve tables underlying
992 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
993 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
994 are judged against seed-level variability, not against this single repeat.

995 Appendix C. Automation disclosure and adjudication gates

996 Appendix C.1. Automation disclosure

997 The mechanical stages of the analysis pipeline—applying pre-specified nu-
998 meric thresholds to aggregated results, re-computing aggregates, re-verifying
999 evaluation fingerprints, and checking each decision against its pre-specified
1000 rule—were carried out by LLM-based coding agents under an orchestration

harness, with every step auditable in the released trail. Manuscript drafting and figure generation were also assisted by such agents (main text, Declaration of Generative AI).

Appendix C.2. Incidents

Three incidents occurred in the original grids, each caught before an erroneous artefact entered a reported number: (1) on the frozen-selectivity grid an automated decision substituted an unspecified criterion for the pre-specified 10 pp threshold; re-applying the original rule gives the reported outcome, “not the primary mechanism”; (2) on the component-ablation grid an automated pass substituted an interval heuristic for the pre-specified threshold, and post-hoc rule files were generated; the true pre-specification predated the grid and was re-applied, and the post-hoc files are retained and marked void; (3) on the graft grid a construction stage was falsely marked as passed; the grid was re-executed from scratch under a corrected task card. Deviations in the later follow-up grids (for example a bootstrap made deterministic after first use, and rounding corrections) are recorded in the repository’s decision memos. Full descriptions, with dates and artefact names, are given in Supplementary Material S1.

Appendix C.3. Human adjudication gates

All numeric adjudication thresholds, every ruling on an incident, the decision to report the incidents, and every interpretive claim in Section 5 were fixed or made by the human authors. No agent could overwrite a protected decision artefact, alter a pre-specified threshold or mark its own output as final.

Appendix C.4. Alignment with Elsevier’s generative-AI policy

In line with Elsevier’s policy, AI tools are not listed as authors and the human authors take full responsibility for the content. We extend the disclosure beyond the writing process to the adjudication pipeline because that is where automation could affect the reported conclusions.

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